B11902010	車以方	Q1

$T(n) = T(n/2) + O(1)$. Unrolling: after k halvings the array has $n/2^k$ elements; we stop when $n/2^k = 1$, i.e. $k = \log_2 n$. Each level does $O(1)$ work, so $T(n) = O(\log n)$. Worst case: the element is absent, and we probe until the range is empty — still $\log_2 n + 1$ probes.

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車以方

Q2

Greedy works. Exchange argument.

B11902010	車以方	Q3

Sort the array, then the LIS is the whole sorted array, so the answer is always n and this runs in O(n log n).

B11902010	車以方	Q4

(pencil smudge) ... midterm of the other course ... (crossed out)

see attached sheet (no sheet attached)